

Rolex Submariner — Hyper-Detailed Physical Description







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The Rolex Submariner is a study in functional precision. Every surface, proportion, transition, and edge exists for a purpose. Despite appearing simple from a distance, it contains hundreds of subtle geometric relationships that give it its unmistakable identity.

Overall Dimensions

The modern Submariner Date (Ref. 126610LN) measures:

- Diameter: **41.00 mm**
- Lug-to-lug: **47.6 mm**
- Thickness: **12.3 mm**
- Lug width: **21.0 mm**
- Bracelet taper:
 - 21 mm at lugs
 - 18 mm at clasp

Weight:

- Approximately **158–162 g** (full bracelet)

Water resistance:

- **300 meters (1,000 ft)**

Crystal height above bezel:

- ~0.4 mm

Caseback protrusion:

- ~1.0 mm below midcase

The Case

Material:

904L Oystersteel (Rolex calls it Oystersteel)

Color:

Neutral silver with an extremely slight warm tint.

Surface finish:

Upper surfaces:

- perfectly linear satin brushing
- grain direction runs exactly parallel to the bracelet

Sides:

- flawless mirror polish
- distortion-free reflection
- no orange-peel texture

Edges:

Every edge is sharply defined.

Transitions:

- brushed top
→ razor edge
→ polished flank

The transition is almost knife-like.

Case Geometry

The Submariner case is known as the **Super Case** (older references) and evolved into a more balanced case on modern models.

The center case resembles a rounded rectangular block.

Characteristics:

Front view:

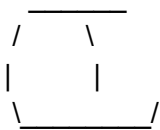
- widest point occurs halfway down the crown guards
- shoulders taper subtly toward the lugs

Top view:

The case isn't perfectly circular.

Instead:

slightly flattened



The curvature is almost imperceptible.

Lugs

Four lugs extend outward.

Upper surface:

- radial brushing

Inner surfaces:

- polished

Outer profile:

Each lug curves downward roughly 9–11°.

Thickness gradually reduces toward the tip.

End profile:

Not square.

More like a rounded trapezoid.

Each lug terminates in drilled holes on vintage models or hidden spring bar holes on modern references.

Crown Guards

One of the most recognizable features.

Located only on the right side.

Shape:

Two asymmetrical shoulders.

Upper guard:

slightly shorter.

Lower guard:
slightly thicker.

Both taper toward the crown.

Viewed from above:

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The guards do not touch the crown.

Tiny equal gaps remain.

Screw-down Crown

Diameter:
~7.0 mm

Height:
~3.6 mm

The crown bears:

- embossed Rolex coronet

Outer edge:

Fine fluting.

Approximately **24 equally spaced grooves**.

These grooves are rounded rather than sharp.

When unscrewed:

- threads become visible
 - crown extends outward in stages
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Bezel

Diameter:
~40.3 mm

Construction:

- stainless steel bezel ring
- ceramic insert (Cerachrom)

Color:
Deep black.

Not matte.

Not glossy.

It possesses a rich satin gloss.

Almost liquid-looking under sunlight.

Bezel Edge

The outer grip consists of approximately **120 machined teeth**.

Each tooth:

- V-shaped
- evenly spaced
- identical depth

Touch:

Aggressive enough for wet fingers.

Not sharp enough to cut skin.

Bezel Rotation

120 clicks.

Each click:

- crisp
- equal force
- almost no play

Sound:

tick

tick

tick

tick

Never mushy.

Rotation:

Counter-clockwise only.

Ceramic Insert

Material:

Zirconium oxide ceramic.

Nearly impossible to scratch.

Laser engraved numerals.

The engraved recesses are coated with platinum using PVD.

Minute Scale

First 15 minutes:

Every minute individually marked.

Remaining minutes:

Five-minute intervals.

Triangle at 12:

Contains a luminous pearl.

Surrounded by polished steel.

Crystal

Material:
Synthetic sapphire.

Thickness:
~1.8–2.0 mm.

Hardness:
9 Mohs.

Edges:

Slightly beveled.

No visible distortion except at extreme viewing angles.

Cyclops (Date Models)

Magnification:

Exactly **2.5×**

Material:
Sapphire.

Perfectly centered.

Bonded seamlessly.

Creates strong distortion outside the date window.

Dial

Diameter:

Approximately 30 mm.

Construction:

Brass base.

Several lacquer layers.

Deep black.

No visible grain.

Appears almost infinite.

Dial Finish

Not matte.

Not glossy.

Micro-satin lacquer.

Light behaves softly.

No sparkle.

Hour Markers

Applied indices.

White gold surrounds.

Filled with Chromalight.

Shapes:

12:
Triangle

3:
Rectangle

6:
Rectangle

9:
Rectangle

Others:
Perfect circles.

Marker Dimensions

Circle markers:

~3.2 mm diameter.

Rectangles:

~5.2 mm long.

Triangle:

~5 mm base.

Each marker stands approximately 0.5 mm above the dial.

Hands

Mercedes Hour Hand

Length:
Approximately 8.5 mm.

Shape:

Rectangular stem.

Circular "Mercedes" symbol.

Three equally sized sectors.

Filled with lume.

Minute Hand

Long sword shape.

Length:

~13.5 mm.

Tip reaches minute track.

Seconds Hand

Thin polished needle.

Circular lume pip.

Tiny pointed counterweight.

Printing

All text uses Rolex's proprietary serif typography.

Extremely crisp.

Pad printed.

No ink bleeding.

Letter spacing is exceptionally consistent.

Rehaut

The inner ring between the dial and crystal.

Mirror polished.

Laser engraved:

ROLEX ROLEX ROLEX ROLEX

Repeated continuously.

At 6 o'clock:

Unique serial number.

Movement (Not Visible)

Calibre 3235.

Diameter:

28.5 mm

Thickness:

6.0 mm

Jewels:

31

Frequency:

28,800 vibrations/hour

Power reserve:

70 hours

Rotor:

18k gold, bidirectional winding.

Accuracy:

-2/+2 seconds/day after casing.

Bracelet

Oyster Bracelet

Three-link construction.

Center links:
Polished sides.

Top:
Linear brushing.

Outer links:
Brushed.

Link Geometry

Each link is:

Slightly arched.

Not flat.

Top radius:
Approximately 0.8 mm.

Edges:
Chamfered microscopically.

End Links

Solid.

Perfectly integrated into the case.

No visible gap.

Clasp

Oysterlock safety clasp.

Outer finish:
Brushed.

Sides:
Polished.

Engraved Rolex coronet.

GlideLock System

Allows approximately **20 mm** of adjustment.

Increment:
~2 mm per click.

Entire mechanism hidden.

Light Behavior

Under direct sunlight:

- polished sides become mirror-like
 - brushed surfaces scatter light uniformly
 - ceramic bezel develops a deep, liquid-black appearance
 - sapphire crystal nearly disappears due to anti-reflective treatment on the underside of the Cyclops
 - white-gold markers catch light with crisp highlights
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Tactile Characteristics

Running a finger across the watch:

- bezel teeth feel evenly machined
 - polished case sides are glass-smooth
 - brushed surfaces have a fine, silky texture
 - crown threads engage with a precise, mechanical resistance
 - clasp closes with a firm, reassuring snap
 - bracelet articulates fluidly with almost no lateral play
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Proportional Ratios

- Bezel outer diameter : case diameter $\approx 0.983 : 1$
- Dial diameter : case diameter $\approx 0.73 : 1$
- Crystal visible area : bezel opening $\approx 95\%$
- Bracelet width : case width $\approx 0.51 : 1$
- Thickness : diameter $\approx 0.30 : 1$

These proportions are a major reason the Submariner appears balanced and timeless: the wide bezel frames the dial without overwhelming it, the lugs are compact enough to wear comfortably on many wrist sizes, and the bracelet integrates seamlessly into the case, creating a continuous visual flow from end to end.